Least Squares Regression (Part 1)

Recommended Online Resource

- Least Squares Regression, Pseudoinverse (Steve Brunton)
- Least Squares Regression and SVD (Steve Brunton)

Example Problem

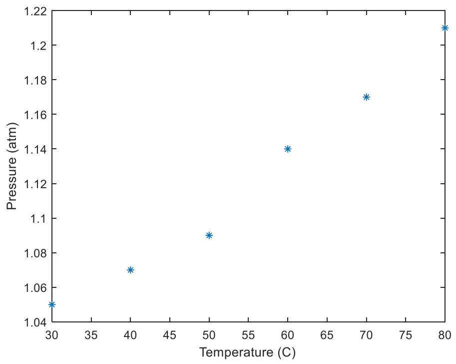
Use linear least-squares regression to predict a value

Problem Statement

An experiment is conducted that measures the pressure of a gas by heating it in a closed chamber. Predict the pressure if the temperature is increased to 150C.

The experiment data is given as

T(°C)	30	40	50	60	70	80
P(atm)	1.05	1.07	1.09	1.14	1.17	1.21



Formulation of the Solution

Based on Charles's law for ideal gas shows a linear relationship of

P=kT, where k is a constant.

Find the best fit curve of P=a0+a1(T)

Solving with NM (MATLAB)

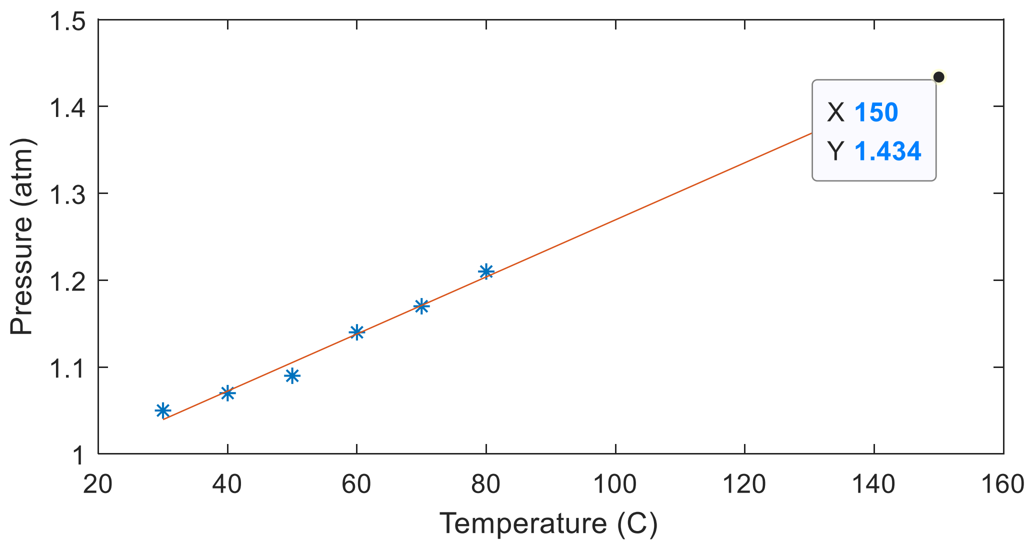
Use MATLAB polyfit(x,y,n) or curve-fitting tool to solve

See MATLAB demo

```
close all, clear all
T=[30  40     50     60     70     80];
P=[1.05       1.07   1.09   1.14   1.17   1.21];
figure(1)
plot(T,P, '*')
xlabel('Temperature (C)')
ylabel('Pressure (atm)')
```

```
a=polyfit(T,P,1)
```

```
t=30:10:150;
yhat=a(2)+a(1).*t;
figure(2)
plot(T,P, '*')
hold on
plot(t,yhat)
xlabel('Temperature (C)')
ylabel('Pressure (atm)')
```

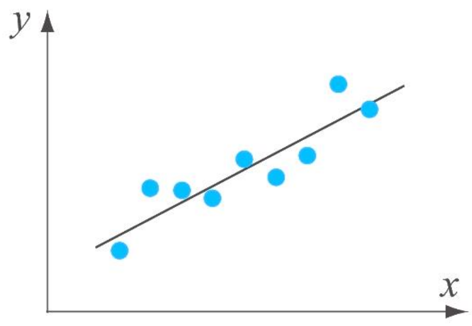


Lecture Objective

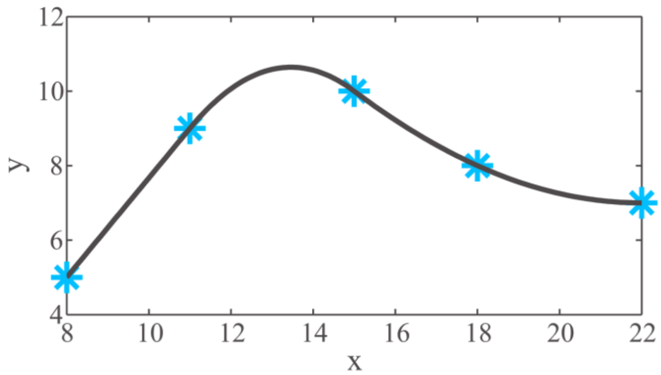
This lecture will cover how to process curve fitting of

- Linear equation
- Quadratic or higher order polynomial
- Other non-linear equation by writing in the linear form

Introduction



Curve-fitting

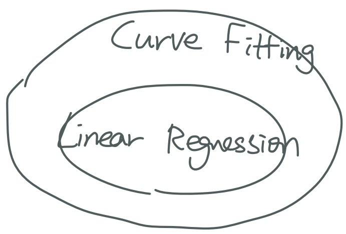


Interpolation

Curve Fitting vs. Interpolation

Curve fitting is finding a function that best fits the overall data points. The function may not cross data points. It is also used to predict a value beyond the given data set range.

Interpolation is used for estimating a value between the known data points. It is done by first determining a polynomial that passes a set of two or more data points: i.e, nth order of polynomial for (n+1) data sets. For a large number of data points used, different low order polynomials are used to interpolate the intervals (spline interpolation)

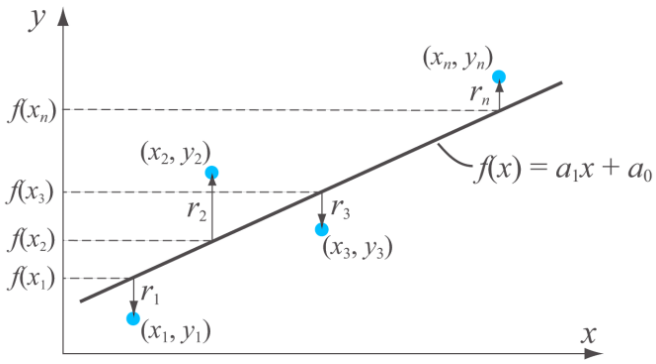


Curve Fitting a Linear Equation

Curve Fitting with a line (first degree polynomial) is finding the best parameters (a_0, a_1) for the linear equation

$$f(x) = \hat{y} = a_0 + a_1 x$$

that gives the *least* total error when a set of data (x_i,y_i) are given.



If there are only two data points, it is simple to derive the line that passes the points. However, when a multiple data points are provided, the best fit line <u>may not </u>pass all the points.

When the number of data points (m) are more than the number parameters (n=2; a_0, a_1), then the system is overdetermined.

- Generally an *inconsistent* system.
- Example:

$$a_0 + a_1 x_k = y_k \text{ ,} \qquad \text{k=1 to m}$$

$$\begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$$

Az=y, where **A**(m x n) , m>(n=2) usually inconsistent

If a given data point (x_k, y_k) is not on the curve fit line of $f(x) = a_0 + a_1 x$

then the actual data point is off from the curve-fit line by the residual error (r_k)

$$y_k = a_1 x_k + a_0 + r_k$$

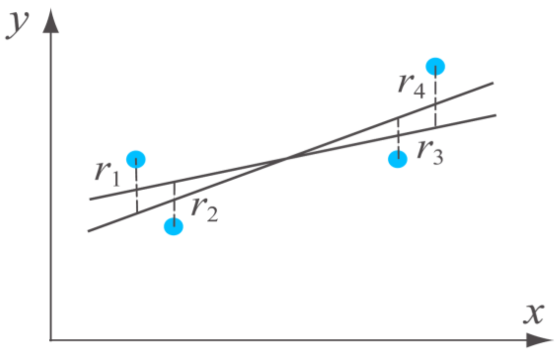
$$r_k = y_k - f(x_k) = y_k - (a_0 + a_1 x_k)$$

Thus, we want to find the *best* curve-fit line such that it gives the *least* total residual error.

Measuring the goodness of the fitting

How can we represent the overall goodness of fitting with one scalar number?
The total residual error (E) can be calculated by different approaches:

- $E = \sum_{i=1}^{m} r_i = \sum_{i=1}^{m} \left(y_i - f(x_i) \right)$, it can give total error E=0
- $E = \sum_{i=1}^{m} |r_i| = \sum_{i=1}^{m} \left(|y_i - f(x_i)| \right)$, the parameters may not be unique



- $E = \sum_{i=1}^{m} r_i^2 = \sum_{i=1}^{m} \left(y_i - f(x_i) \right)^2$ gives unique line function (we will use this)
- $E = \max |r_i|$

Additional notes

For *m* datasets,

The average of total residual error is: $E_{avg} = \dfrac{E}{m} = \dfrac{1}{m}\sum_{i=1}^{m}\left(y_i - f(x_i)\right)^2$

- Variance of the error curve $\mathrm{var}(r) = E[(r-\mu_r)^2]$

Root mean square error(RMSE) $$RMSE = \sqrt{\frac{1}{m}\sum_{i=1}^{m} r_i^2} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(y_i - f(x_i)\right)^2}$$
$$= \sqrt{E_{avg}}$$

Linear Least Square Regression

From a dataset of (x_k, y_k), k=1 to m, fitting a linear equation

- Curve-fit line: $f(x) = \hat{y} = a_0 + a_1 x$
- Residual: $r_k = y_k - f(x_k) = y_k - (a_0 + a_1 x_k)$
- Total error $E = \sum_{k=1}^{m} r_k^2 = \sum_{k=1}^{m} \left(y_k - f(x_k)\right)^2 = \sum_{k=1}^{m} \left(y_k - (a_0 + a_1 x_k)\right)^2$

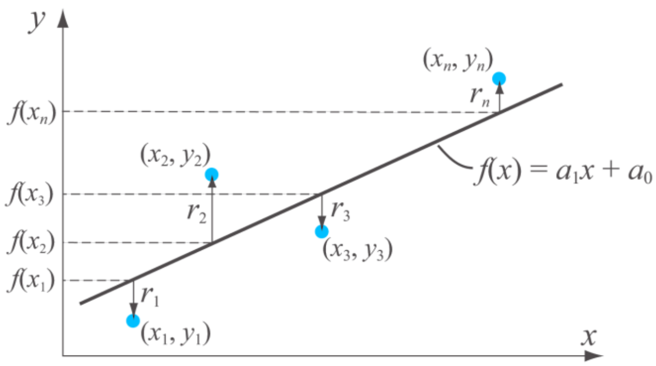
From a dataset of (x_k, y_k), k=1 to m, the residual error from a curve-fit line can be expressed as

$$r_k = y_k - (a_0 + a_1 x_k)$$

Expressing in a system of linear equations

$$\begin{bmatrix} r_1 \\ \vdots \\ r_m \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} - \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$$

$$\mathbf{r} = \mathbf{y} - \mathbf{A}\mathbf{z}$$



Find the best curve-fit line $f(x) = \hat{y} = a_0 + a_1 x$ that minimizes the total error

$$E = \sum_{k=1}^{m} r_k^2 = \sum_{k=1}^{m} \left(y_k - f(x_k)\right)^2 = \sum_{k=1}^{m} \left(y_k - (a_0 + a_1 x_k)\right)^2$$

For the actual data $\mathbf{y} = \mathbf{A}\mathbf{z} + \mathbf{r}$ find the best curve fit line parameter **z***=[a_0^*, a_1^*],

$$\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} + \begin{bmatrix} r_1 \\ \vdots \\ r_m \end{bmatrix} \quad \rightarrow \quad \hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{z}} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix} \begin{bmatrix} a^*_0 \\ a^*_1 \end{bmatrix}$$

There are several approaches for least square regression method.

Approach 1:

Minimize residual errors analytically $\frac{\partial E}{\partial a_i} = 0$ with respect to parameters a_1, a_0

Approach 2:

Using Normal equation $\hat{\mathbf{z}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}$ for $\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{z}}$

- Parameters for the best fit line: $\hat{\mathbf{z}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}$
- Points on the best fit line: $\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{z}} = \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}$

Approach 3:

Using SVD/Pseudoinverse

- Pseudoinverse of A: $\mathbf{A}^+ = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^T$
- Then, finding the best parameters for unsolvable $\mathbf{y} = \mathbf{A}\hat{\mathbf{z}}$ **is**

$$\begin{aligned}\hat{\mathbf{z}} &= \mathbf{A}^+\mathbf{y} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y} \\ &= \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^T\mathbf{y}\end{aligned}$$

- Points on the best line are

$$\begin{aligned}\hat{\mathbf{y}} &= \mathbf{A}\hat{\mathbf{z}} = \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y} \\ &= (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T)\mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^T\mathbf{y} = \mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^+\mathbf{U}^T\mathbf{y}\end{aligned}$$

Approach 4:

QR decomposition: **A=QR**

$$\mathbf{A}\mathbf{z} = \mathbf{y} \rightarrow \mathbf{Q}\mathbf{R}\mathbf{z} = \mathbf{y} \rightarrow \mathbf{R}\hat{\mathbf{z}} = \mathbf{Q}^T\mathbf{y}$$

- **Q**:mxn, **R**:nxn. Note that $\mathbf{Q}^T\mathbf{Q}=\mathbf{I}$(nxn)
- Only back-substitution is used

$$E=\sum_{k=1}^{m}\left(y_k-(a_0+a_1x_k)\right)^2$$

Approach 1: Minimize $\frac{\partial E}{\partial a_i}=0$ with respect to parameters a_1, a_0.

Let there are m number of datasets.

The total residual error is at its minimum with parameters a_1, a_0 for the condition of

$$\frac{\partial E}{\partial a_0}=\sum_{k=1}^{m}2\left(y_k-(a_0+a_1x_k)\right)\cdot(-1)=-2\sum_{k=1}^{m}\left(y_k-(a_0+a_1x_k)\right)$$

$$\frac{\partial E}{\partial a_1}=\sum_{k=1}^{m}2\left(y_k-(a_0+a_1x_k)\right)(-x_k)=-2\sum_{k=1}^{m}x_k\left(y_k-(a_0+a_1x_k)\right)$$

The parameter a_1, a_0 can be solved as a linear system:

$$\left(\sum_{k=1}^{m}1\right)a_0+\left(\sum_{k=1}^{m}x_k\right)a_1=\sum_{k=1}^{m}y_k$$

$$\left(\sum_{k=1}^{m}x_i\right)a_0+\left(\sum_{k=1}^{m}x_k^{\,2}\right)a_1=\sum_{k=1}^{m}x_ky_k$$

$$\Rightarrow\begin{bmatrix}m & S_x\\ S_x & S_{xx}\end{bmatrix}\begin{bmatrix}a_0\\ a_1\end{bmatrix}=\begin{bmatrix}S_y\\ S_{xy}\end{bmatrix}$$

$$\begin{bmatrix}a_0\\ a_1\end{bmatrix}=\begin{bmatrix}m & S_x\\ S_x & S_{xx}\end{bmatrix}^{-1}\begin{bmatrix}S_y\\ S_{xy}\end{bmatrix}$$

$$\begin{bmatrix}a_0\\ a_1\end{bmatrix}=\frac{1}{mS_{xx}-S_xS_x}\cdot\begin{bmatrix}S_{xx} & -S_x\\ -S_x & m\end{bmatrix}\begin{bmatrix}S_x\\ S_{xy}\end{bmatrix}\qquad \hat{\mathbf{z}}=(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}\left(\mathbf{A}^{\mathrm{T}}\mathbf{y}\right)$$

The parameters can be more simply calculated as

$$a_0=\frac{S_{xx}S_y-S_xS_{xy}}{mS_{xx}-S_xS_x},\qquad a_1=\frac{mS_{xy}-S_xS_x}{mS_{xx}-S_xS_x}$$

where S_{xx}, S_x, S_y, S_{xy} are obtained easily with the data points (x_i, y_i)

Approach 2: Using Normal equation $\hat{\mathbf{z}} = (\mathbf{A}^{\mathbf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathbf{T}}\mathbf{y}$

Let there are *n* number of datasets.

With the given data points(x_i, y_i), residuals and total error can be expressed as

$$\mathbf{r} = \mathbf{y} - \mathbf{Az}$$

$$\begin{bmatrix} r_1 \\ \vdots \\ r_m \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} - \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$$

$$E = \sum_{k=1}^{m} r_k^{\,2} = \mathbf{r}^{\mathbf{T}}\mathbf{r} = \boxed{r \cdot r = (y - Az)^T(y - Az)}$$

The best fit line should minimize the total error E with respect to the parameter vector **z=[**a0, a1**]**

$$\frac{\partial E}{\partial \mathbf{z}} = \mathbf{0} \qquad // \quad E = \mathbf{r}^{\mathbf{T}}\mathbf{r} = (\mathbf{y} - \mathbf{Az})^T(\mathbf{y} - \mathbf{Az}).$$

The derivative of *E (scalar)* wr.t. **z** (vector) becomes

$$\frac{\partial E}{\partial \mathbf{z}} = 0 = \frac{\partial}{\partial \mathbf{z}}\left((\mathbf{y} - \mathbf{Az})^T(\mathbf{y} - \mathbf{Az})\right)$$

$$= \frac{\partial}{\partial \mathbf{z}}\left((\mathbf{y} - \mathbf{Az})^T\mathbf{y} - (\mathbf{y} - \mathbf{Az})^T\mathbf{Az}\right)$$

$$= -\frac{\partial}{\partial \mathbf{z}}(\mathbf{z}^{\mathbf{T}}\mathbf{A}^{\mathbf{T}}\mathbf{y}) - \frac{\partial}{\partial \mathbf{z}}(\mathbf{y}^{\mathbf{T}}\mathbf{Az}) + \frac{\partial}{\partial \mathbf{z}}(\mathbf{z}^{\mathbf{T}}\mathbf{A}^{\mathbf{T}}\mathbf{Az})$$

$$= -(\mathbf{A}^{\mathbf{T}}\mathbf{y})^{\mathbf{T}} - (\mathbf{y}^{\mathbf{T}}\mathbf{A}) + \mathbf{z}^{\mathbf{T}}(\mathbf{A}^{\mathbf{T}}\mathbf{A} + \mathbf{A}^{\mathbf{T}}\mathbf{A})$$

$$= -2(\mathbf{y}^{\mathbf{T}}\mathbf{A}) + 2(\mathbf{z}^{\mathbf{T}}\mathbf{A}^{\mathbf{T}}\mathbf{A})$$

$$\Rightarrow \quad (\mathbf{z}^{\mathbf{T}}\mathbf{A}^{\mathbf{T}}\mathbf{A}) = (\mathbf{y}^{\mathbf{T}}\mathbf{A})$$

Applying transpose on both sides yields

$$(\mathbf{A}^{\mathbf{T}}\mathbf{A})\mathbf{z} = (\mathbf{A}^{\mathbf{T}}\mathbf{y})$$

$$\mathbf{z} = \boxed{(A^TA)^{-1}(A^Ty)}$$

Derivative of a scalar w.r.t vector

$$\frac{\partial E}{\partial \mathbf{z}} = \left[\frac{\partial E}{\partial z_1}, \quad \frac{\partial E}{\partial z_2}, \quad \cdots\right]$$

$$\frac{\partial}{\partial \mathbf{z}}(\mathbf{z}^{\mathbf{T}}\mathbf{y}) = \mathbf{y}^{\mathbf{T}}$$

$$\frac{\partial}{\partial \mathbf{z}}(\mathbf{y}^{\mathbf{T}}\mathbf{z}) = \mathbf{y}^{\mathbf{T}}$$

$$\frac{\partial}{\partial \mathbf{z}}(\mathbf{z}^{\mathbf{T}}\mathbf{Az}) = \mathbf{z}^{\mathbf{T}}(\mathbf{A}^{\mathbf{T}} + \mathbf{A})$$

Note: $\frac{\partial E}{\partial \mathbf{z}} = 0 = \mathbf{A}^{\mathbf{T}}(\mathbf{y} - \mathbf{Az})$

*This is finding the **z** vector that makes the **r** perpendicular to the column space of A*

The points **on** the best fit line are $(x_i, \hat{y}_i)$:

$$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{z}} = A(A^TA)^{-1}(A^Ty) \quad \mathbf{A} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix}$$

Summary: Normal equation for linear least square regression

The parameters for the best fit line are

$$\hat{\mathbf{z}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}$$

The points on the best fit line are

$$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{z}} = \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}$$

Q. How can we check if $\frac{\partial E}{\partial \mathbf{z}} = 0$ is at the minimum point? Can it be the maximum point of E?

A quadratic curve is at the minimum if $f'(x)=0$ & $f''(x)>0$

$$\frac{\partial E}{\partial z} = -2(y^TA) + 2(z^TA^TA), \quad \frac{\partial^2 E}{\partial z^2} = \frac{\partial}{\partial z}\left(-2(y^TA) + 2(z^TA^TA)\right)$$

$$= 2(A^TA) > 0$$

Therefore, $\frac{\partial E}{\partial z} = 0$ is at the minimum point.

Note:

- $\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$ is the projection matrix. What is the definition of projection matrix?
- $(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$ is the pseudoinverse of **A** // $Az=y$, $A^TAz=A^Ty$, $z=(A^TA)^{-1}A^Ty$

- What are the elements of $(\mathbf{A}^{\mathbf{T}}\mathbf{A})$?

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \underbrace{\frac{1}{mS_{xx} - S_x S_x} \begin{bmatrix} S_{xx} & -S_x \\ -S_x & m \end{bmatrix}}_{\substack{\shortparallel \\ (\mathbf{A}^T\mathbf{A})^{-1}}} \begin{bmatrix} S_y \\ S_{xy} \end{bmatrix} \quad \text{vs} \quad \hat{\mathbf{z}} = (\mathbf{A}^{\mathbf{T}}\mathbf{A})^{-1}\left(\mathbf{A}^{\mathbf{T}}\mathbf{y}\right)$$

$$\hat{\mathbf{z}} = (\mathbf{A}^{\mathbf{T}}\mathbf{A})^{-1}\left(\mathbf{A}^{\mathbf{T}}\mathbf{y}\right)$$

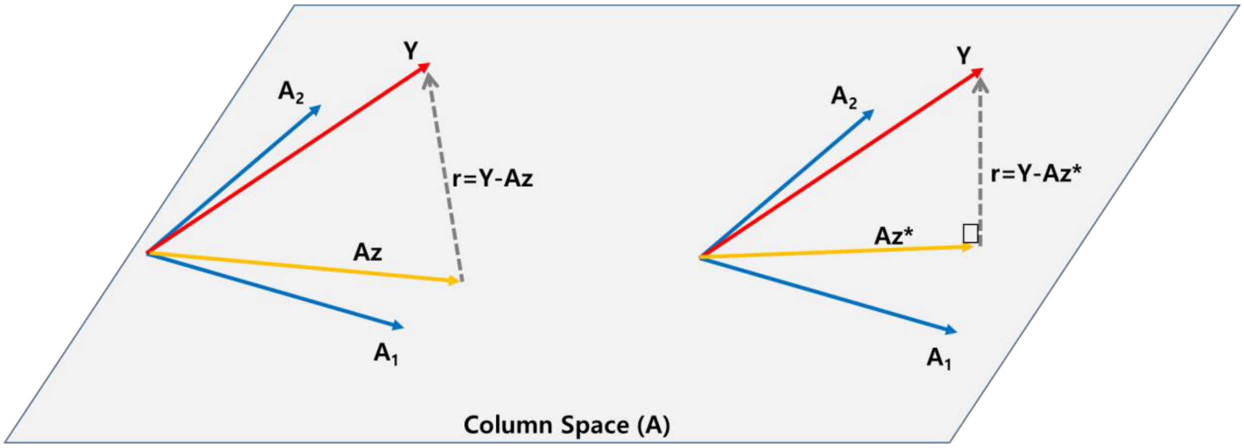
$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} m & S_x \\ S_x & S_x S_x \end{bmatrix}^{-1} \begin{bmatrix} S_y \\ S_{xy} \end{bmatrix}$$

$$= \begin{bmatrix} m & \sum_{k=1}^{m} x_k \\ \sum_{k=1}^{m} x_k & \sum_{k=1}^{m} x_k^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{k=1}^{m} y_k \\ \sum_{k=1}^{m} x_k y_k \end{bmatrix}$$

Geometric Intuition

$$\frac{\partial E}{\partial \mathbf{z}} = 0 = \mathbf{A}^{\mathrm{T}}(\mathbf{y} - \mathbf{Az})$$

Finding the parameter **z*** that minimizes the total error is finding the **z** vector that makes the **r** error perpendicular to the column space of **A**



The column space of **A** is all the linear combinations of the column vectors of **A**.

For $\mathbf{A} = [\mathbf{A}_1\ \mathbf{A}_2\ \mathbf{A}_3]$, and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, column space is the span by column vectors of A

$$\mathbf{Ax} = x_1\mathbf{A}_1 + x_2\mathbf{A}_2 + x_3\mathbf{A}_3 \quad ,$$

For Linear regression , **A** (nx2),the column space of A

$$\mathbf{Az} = z_1\mathbf{A}_1 + z_2\mathbf{A}_2 \quad \text{with} \quad \mathbf{A} = [\mathbf{A}_1\ \mathbf{A}_2\ ], \text{ and } \quad \mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

This is the *plane* defined by the two vectors $\mathbf{A}_1$, and $\mathbf{A}_2$.

The total error is the distance from the planar column space of **Az** and **y**.

$$E = \|\mathbf{y} - \mathbf{Az}\|^2 = \|\mathbf{r}\|^2$$

If the data point **y** lies on the column space of **A,** then, there is no error for

Az=y, E**=norm(Az-y)=**0

The error is at minimum if the error vector **r** is perpendicular to the column space of **A**(or the plane). Thus the vector **z*** is such that it

- makes **Az*** vector on the plane to be perpendicular to error vector **r**
- or it is the projected vector of **y** on to the column space of **A**

(Exercise) Pseudo code of Linear Least Square Regression

data points $(x_1, y_1), \cdots, (x_m, y_m)$ are given.

Objective : obtain $\hat{z} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$

$$a_0 = \frac{S_{xx}S_y - S_xS_{xy}}{mS_{xx} - S_xS_x}, \quad a_1 = \frac{mS_{xy} - S_xS_y}{mS_{xx} - S_xS_x} \quad \left(\text{where, } \sum^m x_k = S_x, \ \sum^m y_k = S_y, \ \sum^m x_k^2 = S_{xx}, \ \sum^m x_k y_k = S_{xy} \right)$$

```
m = # of data points
x[m] = { x1, ... , xm }
y[m] = { y1, ... , ym }

for i = 1 to m
    Sx += x[i]
    Sy += y[i]
    Sxx += x[i]^2
    Sxy += x[i] * y[i]
end i

a0 = (Sxx * Sy - Sx * Sxy) / (m * Sxx - Sx * Sx) ;

a1 = (m * Sxy - Sx * Sy) / (m * Sxx - Sx * Sx) ;
```

(Extra) How SVD (Singular Value Decomposition) can be applied for Linear Regression?

Summary

Find the linear equation that minimizes the total residual error, where residual is

$$r_k = y_k - \hat{y}_k = y_k - \left(a_0 + a_1 x_k\right)$$

Solution:

$$\text{minimize } \left\|\mathbf{y} - \hat{\mathbf{y}}\right\|^2 \left(= \left\|\mathbf{y} - \mathbf{Az}\right\|^2\right)$$

$$\hat{\mathbf{z}} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} m & \sum_{k=1}^{m} x_k \\ \sum_{k=1}^{m} x_k & \sum_{k=1}^{m} {x_k}^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{k=1}^{m} y_k \\ \sum_{k=1}^{m} x_k y_k \end{bmatrix} = (\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathrm{T}}\mathbf{y}\,, \quad \text{where } \mathbf{A} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix}$$

Simple solution:

$$a_0 = \frac{S_{xx}S_y - S_x S_{xy}}{mS_{xx} - S_x S_x}\,, \quad a_1 = \frac{mS_{xy} - S_x S_y}{mS_{xx} - S_x S_x}$$